

Pre-Registration BIOL530H Mini-Project

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Overview

Research questions or hypotheses

I hypothesized that patient body weight significantly effects the elimination half-life of theophylline, potentially due to weight-dependent changes in the volume of distribution or physiological clearance rates. Null Hypothesis: There is no linear relationship between a patient's body weight and the elimination half-life of theophylline. Alternative Hypothesis: There is a linear relationship between a patient's body weight and the elimination half-life of theophylline.

Foreknowledge of data or evidence

Analyses in this plan have been conducted already. At least some of the analyses described in this analysis plan have been conducted by the authors making this a retrospective registration.

Explanation of foreknowledge and managing unintended influences

This data set is publicly available and built into R studio, its general pharmacokinetic properties are already well documented in existing literature. To prevent bias in my interpretation of the data, I deliberately did not research the drug prior to my analysis. This approach was chosen to mimic a true exploratory study where the expected outcomes are not known in advance.

Research Design

Study type

Non-randomized study: Does not include random assignment of subjects to treatments or conditions. This usually includes correlational studies, observational studies, surveys, natural experiments, real-world evidence studies, regression discontinuity designs, and descriptive studies.

Intention for causal interpretation

Indirect inference on causal relationship(s): This study is intended to examine a potential causal relationship but is not designed to isolate the specific causal effect. This work may inform causal inference indirectly or advance insight toward future estimation of a causal relationship.

Blinding of experimental treatments

No blinding is involved.

Study design

I am using existing data (the Theoph dataset), so I did not run an experiment or assign treatments to patients. The original researchers measured 12 patients multiple times over 25 hours, making it a repeated-measures design. However, my specific study just compares those different patients to

each other, which is a between-subjects design. I am simply comparing each patient's weight to their calculated drug half-life to see if they are correlated. Because I am just doing math on an existing dataset, complex experimental setups like randomization, blinding, or counterbalancing do not apply.

Sampling

Data collection procedures

This study relies entirely on a pre-existing, publicly available dataset (the Theoph dataset), which is included in R Studio. No new data will be collected, so no human subjects will be recruited or sampled for this analysis. The original data was collected by Dr. Robert Upton around 1980 and consists of 12 subjects who received a single oral dose of the anti-asthmatic drug theophylline. The original publication is unclear and not cited (if it exists). Data preparation will involve loading the dataset directly from R studio and processing the raw data for relevant variables such as subject weight, time, and concentration. All 12 subject will be included in all analyses, but specific time points representing the absorption phase will be excluded during data preparation to isolate the elimination phase—this is required for the half-life calculation for each subject.

Boeckmann, A. J., Sheiner, L. B. and Beal, S. L. (1994), NONMEM Users Guide: Part V, NONMEM Project Group, University of California, San Francisco. However, Dr. Robert A. Upton, is not credited in this citation, nor is his original work cited.

Sample size

The total sample size for this secondary analysis is $N = 12$ Subjects. The primary unit of analysis is the individual subject. The dataset was originally 132 observations, but these observations were used to calculate just one half-life number for each person. Therefore, the final simple linear regression model will be conducted on exactly 12 paired data points (12 weights and 12 calculated half-lives). I won't be splitting these 12 people into groups so sub-sample sizes, nesting, and cross-condition comparisons are not applicable.

Sample size rationale

Since the sample size of $N = 12$ is determined by the constraint of pre-existing data, no *a priori* statistical power analyses were conducted to obtain the optimal sample size. All subjects were included in the analyses to maximize available power.

Starting and stopping rules

No new data is being generated, and the entirety of the dataset is being used, so starting and stopping rules are not applicable.

Variables

Manipulated variables

None—this is a secondary analysis of pre-existing data.

Measured variables

The predictor variable is patient body weight measured in kilograms. The outcome measure is the elimination half-life measured in hours (derived from the raw data). Variables used for calculation and subsetting are Subject, Time, and Concentration. Concentration was converted to a natural log to calculate the elimination rate constant, which was used to find the half-life. Further subsetting of subject and weight from the raw Theoph data set was conducted and combined with the half-life calculations for each subject.

Indices

Natural log of Concentration. Calculated by taking the $\ln(\text{conc})$. Concentration of theophylline was log-transformed to linearize the elimination phase. Elimination rate constant (Kel). Derived by determining the slope from a simple linear regression of $\ln(\text{conc})$ plotted against Time during the elimination phase for each subject. Calculated by taking the absolute value of slope $Kel = |\text{Slope}|$ Elimination Half-life. Derived for each subject using their individual calculated elimination rate constant (Kel) using formula: $t_{1/2} = \ln(2)/Kel$

Analysis Plan

Statistical models

To test the hypothesis that patient body weight influences the elimination half-life of theophylline, a simple linear regression (Ordinary Least Squares) model was used. The model assumes linearity. The model's primary outputs of interest are the p-value (significance threshold at $\alpha = 0.05$) and the R-squared value to determine the effect size of weight on half-life.

Transformations

Concentration of theophylline in the blood was log transformed ($\ln(\text{conc})$) to linearize the elimination phase of the curve. This transformation was necessary to calculate the elimination rate constant (Kel) and derive the elimination half-life, which served as a primary outcome variable. No centering or scaling was performed on the weight variable.

Inference criteria

The primary criterion for statistical significance is the p-value associated with the weight variable in the linear regression model. The standard alpha level of 0.05 will be used as the cut-off to determine statistical significance. Analysis utilizes a two-tailed test. Model evaluates whether body weight significantly influences half-life, rather than predicting a unidirectional change. The multiple R-squared value will be used as a descriptive index of model fit to report the proportion of variance in theophylline half-life that is explained by body weight. No adjustments for multiple comparisons are required.

Data inclusion and exclusion

All 12 subjects present in the existing theophylline data set were included in the final linear regression analysis. I couldn't use blood samples from the first couple of hours when the drug was still absorbing into their system, so I intentionally excluded those early data points to look at only

the data points where the drug was leaving their body. No data points or subjects were excluded on the basis of being statistical outliers.

Missing data

There were no missing data points for any of the 12 subjects across the measured time points, concentrations, or bodyweights.